

MACAU INTL, Macao

WMO: 450110

Lat: 22.1595N

Lon: 113.5680E

Elev: 374

StdP: 14.50

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.2	48.4	31.8	26.5	53.2	35.8	31.2	53.5	27.3	50.0	25.2	51.0	18.3	0	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	91.0	80.6	89.6	80.5	88.2	80.4	82.9	87.1	82.3	86.2	81.7	85.7	8.4	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.1	169.7	86.1	80.9	162.9	85.3	80.6	161.4	85.1	47.4	87.3	46.7	87.5	46.0	86.1	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	22.1	19.8	43.0	95.3	2.9	1.9	40.9	96.7	39.2	97.8	37.6	98.9	35.4	100.3
			38.3	84.4	2.9	1.2	36.2	85.3	34.5	85.9	32.9	86.6	30.7	87.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.7	60.0	61.4	65.9	72.9	79.0	82.5	83.7	83.5	82.2	78.4	71.4	63.2
	DBStd	9.80	5.91	6.75	5.31	4.82	3.59	2.80	2.48	2.51	2.91	3.44	5.10	5.61
	HDD50	10	4	5	0	0	0	0	0	0	0	0	0	1
	HDD65	476	173	132	54	6	0	0	0	0	0	0	12	99
	CDD50	8674	313	325	493	687	899	976	1045	1039	965	879	643	410
	CDD65	3661	16	32	81	242	434	526	580	574	515	414	204	43
	CDH74	34558	5	11	66	957	3707	6053	7109	6972	5729	3283	638	28
	CDH80	10882	0	0	1	83	778	2089	2790	2689	1856	557	38	1
Wind	WSAvg	10.0	10.5	9.8	9.1	9.3	9.5	9.7	9.6	8.7	9.6	10.7	11.2	11.6
Precipitation	PrecAvg	76.6	1.3	1.5	2.7	6.1	12.5	14.5	10.0	12.4	8.5	3.9	1.7	1.3
	PrecMax	113.5	10.9	8.6	9.1	23.0	38.4	47.4	33.4	29.0	23.7	15.0	9.6	5.9
	PrecMin	38.7	0.0	0.0	0.3	0.2	1.7	3.9	1.3	2.9	0.4	0.0	0.0	0.0
	PrecStd	19.6	1.8	1.7	2.1	5.0	8.1	8.3	6.6	6.4	6.1	4.3	1.9	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.0	75.5	78.5	84.6	89.3	90.6	93.8	93.4	91.9	88.1	83.1	77.1
		MCWB	66.6	69.0	73.6	78.0	80.7	80.9	82.4	80.6	78.9	77.6	74.1	69.3
	2%	DB	70.5	73.1	75.6	82.3	87.4	89.3	90.4	90.9	89.6	86.1	80.8	73.8
		MCWB	65.0	68.6	71.7	77.4	80.1	81.1	81.3	80.6	79.0	76.6	73.2	68.4
	5%	DB	68.3	71.1	73.7	80.5	85.6	87.9	89.2	89.4	87.9	84.4	79.0	71.8
		MCWB	63.8	67.5	70.5	76.3	79.9	81.0	81.0	80.6	79.2	75.9	72.5	66.5
	10%	DB	66.6	69.0	71.9	78.8	84.0	86.4	87.8	87.9	86.4	82.7	77.3	70.0
		MCWB	62.6	65.5	68.9	75.6	79.2	80.5	80.8	80.4	78.5	74.9	71.7	64.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.9	71.6	75.0	79.6	82.5	83.5	84.1	83.8	82.7	80.1	76.7	72.6
		MCDB	71.8	73.3	77.0	82.4	86.1	87.5	89.7	88.8	87.1	84.7	80.1	74.8
	2%	WB	66.7	69.7	73.1	78.3	81.4	82.6	82.8	82.6	81.6	79.0	75.5	70.1
		MCDB	69.4	72.1	74.9	80.9	84.7	86.1	87.2	87.0	85.9	83.2	78.8	73.0
	5%	WB	65.1	68.0	71.2	77.4	80.5	81.8	82.2	81.8	80.7	78.1	74.4	68.0
		MCDB	67.8	70.4	73.2	79.9	83.7	85.4	86.2	86.2	85.1	82.3	77.8	71.0
	10%	WB	63.6	65.9	69.7	76.1	79.7	81.3	81.6	81.2	79.9	77.1	72.9	65.8
		MCDB	66.3	68.3	71.7	78.4	82.8	85.0	85.7	85.6	84.4	81.5	76.7	69.2
Mean Daily Temperature Range	5% DB	MDBR	9.4	8.4	7.9	7.4	7.6	7.4	8.0	8.5	8.5	8.7	9.2	9.6
		MCDBR	11.3	10.9	9.8	9.0	9.1	8.5	9.6	10.3	10.2	9.8	10.2	11.1
		MCWBR	6.6	6.3	6.1	5.1	4.8	4.3	4.6	5.2	5.3	5.2	5.5	6.5
	5% WB	MCDBR	9.9	9.7	8.7	7.7	7.9	7.7	8.6	9.1	9.0	8.3	8.9	9.6
MCWBR		6.6	6.4	6.3	5.2	4.9	4.5	4.9	5.2	4.9	4.7	5.6	6.4	
Clear-Sky Solar Irradiance	taub	0.593	0.636	0.762	0.734	0.572	0.536	0.521	0.592	0.634	0.696	0.592	0.585	
	taud	1.784	1.725	1.522	1.610	2.004	2.122	2.158	1.938	1.816	1.647	1.847	1.823	
	Ebn at Noon	213	214	194	204	239	246	249	233	220	197	211	208	
	Edh at Noon	64	72	92	85	57	50	49	61	67	77	60	60	
All-Sky Solar Radiation	RadAvg	945	987	1074	1217	1463	1507	1726	1597	1489	1372	1104	989	
	RadStd	158	157	144	185	168	213	183	129	126	128	128	148	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+0.27	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	+0.40	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page